

Problem 1.1 An elastic rod 25 mm in diameter, 200 mm long extends by 0.25 mm under a tensile load of 80 kN. Find the intensity of stress, the strain and the elastic modulus for the material of the rod.

Solution: Diameter of rod $d = 25 \text{ mm}$

Length of the rod $l = 200 \text{ mm}$

Tension $P = 80 \text{ kN}$

Area of the section $A = \frac{\pi d^2}{4} = \frac{\pi \times 25^2}{4} = 490.87 \text{ mm}^2$

Intensity of stress $\sigma = \frac{P}{A} = \frac{80 \times 1000}{490.87} \text{ N/mm}^2$
 $= 162.97 \text{ N/mm}^2$

Strain $= e = \frac{\delta}{L} = \frac{0.25}{200} = 0.00125$

Elastic Modulus $= E = \frac{\sigma}{e} = \frac{162.97}{0.00125}$
 $= 1.3 \times 10^5 \text{ N/mm}^2$

Problem 1.2 A hollow steel column of external diameter 250 mm has to support an axial load of 4000 kN. If the ultimate stress for the steel column is 480 N/mm^2 . Find the internal diameter of the column allowing a factor of safety of 4.

Solution: External diameter $D = 250 \text{ mm}$

Internal diameter $d \text{ mm}$

Load on the column $P = 4000 \times 1000 \text{ N}$

Ultimate stress $= 480 \text{ N/mm}^2$

Factor of safety $= 4$

$\therefore$ Safe stress $\sigma = \frac{480}{4} = 120 \text{ N/mm}^2$

$\therefore$ Sectional area required $A = \frac{P}{\sigma} = \frac{4000 \times 1000}{120} = 33333.33 \text{ mm}^2$

$\therefore \frac{\pi}{4}(250^2 - d^2) = 33333.33$

$\therefore 62500 - d^2 = 21221$

$\therefore d^2 = 29167$

$\therefore d = 170 \text{ mm}$

Problem 1.3 A short hollow cast iron cylinder of wall thickness 10 mm is to carry a compressive load of 600 kN. Determine its outside diameter if the ultimate crushing stress for the material is 540 N/mm². Use a factor of safety of 5.

Solution:

Let D_0 = Outside diameter of the cylinder in mm.

The wall thickness of the cylinder = 10 mm

∴ Inside diameter of the cylinder = $(D_0 - 20)$ mm

$$\begin{aligned}\text{Area of cross-section} &= \frac{\pi}{4} \{ D_0^2 - (D_0 - 20)^2 \} \\ &= \pi (10D_0 - 100) \text{ mm}^2\end{aligned}$$

Crushing load for the column = $\pi (10D_0 - 100) \times 540$ N

Factor of safety = 5

$$\therefore \text{Safe load} = 600 \times 10^3 = \frac{\text{Crushing load}}{\text{Factor of safety}} = \frac{\text{Crushing load}}{\text{Factor of safety}}$$

$$600 \times 10^3 = \frac{\pi (10D_0 - 100) \times 540}{5}$$

$$\therefore D_0 = \frac{600 \times 10^3 \times 5}{\pi \times 540} + 100 = 186.93 \text{ mm}$$

Problem 1.4 Calculate the force required for punching a hole of 20 mm diameter through a mild steel plate 4 mm thick. Take maximum shear strength of mild steel as 320 N/mm². Also find the compressive stress on the punch.

Solution: Let d be the diameter of the hole to be punched, t be the thickness of the plate and τ be the maximum shear stress.

Force required for punching the hole = Shear stress $\times$ Area

$$\begin{aligned}&= \tau \times \pi dt = 320 \times \pi \times 20 \times 4 \\ &= 80384 \text{ N}\end{aligned}$$

Area of the punch in contact with the plate surface, $A = \frac{\pi}{4} d^2$

$$\therefore \text{Compressive stress on the punch, } \sigma_c = \frac{P}{A} = \frac{80384}{\frac{\pi}{4} \times (20)^2} = 256 \text{ N/mm}^2.$$

Problem 1.5 The following data refer to a tensile test conducted on a mild steel bar.

(i) Diameter of the steel bar = 30 mm

(ii) Gauge length = 200 mm

- (iii) Extension at a load of 100 kN = 0.139 mm
 (iv) Load at elastic limit = 250 kN
 (v) Maximum Load = 400 kN
 (vi) Total = 56 mm
 (vii) Diameter of the rod at failure = 22.25 mm

Calculate (a) the young's modulus, (b) The stress at elastic limit, (c) the percentage elongation and (d) The percentage decrease in area.

Solution: Area of the rod $= A = \frac{\pi}{4} (30)^2 \text{ mm}^2 = 706.86 \text{ mm}^2$

(a) Young's Modulus

Stress $= \sigma = \frac{P}{A} = \frac{100 \times 1000}{706.86} \text{ N/mm}^2 = 141.47 \text{ N/mm}^2$

Strain $= e = \frac{dl}{l} = \frac{0.139}{200} = 0.000695$

$\therefore$ Young's Modulus $E = \frac{\sigma}{e} = \frac{141.47}{0.000695} \text{ N/mm}^2 = 2.0355 \times 10^5 \text{ N/mm}^2$

(b) Stress at elastic limit $= \frac{250 \times 10^3}{706.86} = 35.68 \text{ N/mm}^2$

(c) Percentage elongation $= \frac{\text{Increase in Length}}{\text{Original Length}} \times 100\% = \frac{56}{200} \times 100\% = 28\%$

(d) Percentage decrease in area

$$= \frac{\frac{\pi}{4}(d^2 - d'^2)}{\frac{\pi d^2}{4}} \times 100\% = \frac{d^2 - d'^2}{d^2} \times 100\%$$

$$= \frac{30^2 - 22.25^2}{30^2} \times 100\% = 44.99\%$$

Problem 1.6 The following data refer to a mild steel specimen tested in a laboratory:

- (i) Diameter of the specimen = 25 mm
 (ii) Length of the specimen = 300 mm
 (iii) Extension under a load of 15 kN = 0.045 mm
 (iv) Load at yield point = 130 kN
 (v) Maximum load = 210 kN

- (vi) Length of the specimen after failure = 375 mm
 (vii) Neck diameter = 17.75 mm

Determine.

- (i) Young's Modulus (ii) Yield point (iii) Ultimate stress (iv) Percentage of elongation (v) Percentage reduction in area (vi) Safe stress adopting a factor of safety of 2.

Solution: Area of the specimen $= A = \pi \times \frac{25^2}{4} = 490.87 \text{ mm}^2$

At a load of 15 kN

Stress intensity $= \sigma = \frac{15 \times 10^3}{490.87} = 30.56 \text{ N/mm}^2$

Strain corresponding to this load $= \frac{0.045}{300} = 1.5 \times 10^{-4}$

Young's Modulus $E = \frac{30.56}{1.5 \times 10^{-4}} = 2.036 \times 10^5 \text{ N/mm}^2$

Stress at yield point $= \frac{130 \times 10^3}{490.87} = 264.84 \text{ N/mm}^2$

Ultimate stress $= \frac{210 \times 10^3}{490.87} = 427.81 \text{ N/mm}^2$

Percentage elongation $= \frac{375 - 300}{300} \times 100 = 25\%$

Percentage reduction in area $= \frac{d^2 - d'^2}{d^2} \times 100 = \frac{25^2 - 17.75^2}{25^2} \times 100\% = 49.59\%$

Safe stress $= \frac{\text{Yield stress}}{\text{factor of safety}} = \frac{264.84}{2} = 132.42 \text{ N/mm}^2$

Problem 1.7 Find the total elongation of a copper bar as shown in Fig. 1.20 subjected to an axial load of 200 kN. Take $E = 100 \text{ GPa}$.

Solution:

Let δL_1 = Elongation for the portion AB of the bar.

δL_2 = Elongation for the portion BC of the bar.

δL_3 = Elongation for the portion CD of the bar.

δL = Total elongation of the copper bar.

$\therefore L_1, L_2$ and L_3 be the length of the portions AB, BC and CD respectively.

The load $P = 200 \text{ kN} = 200 \times 10^3 \text{ N}$.

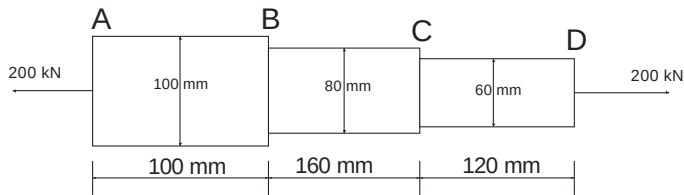


Fig. 1.20

Now
$$\delta L_1 = \frac{PL_1}{A_1 E} = \frac{200 \times 10^3 \times 0.10}{\frac{\pi}{4} \times (0.1)^2 \times 100 \times 10^9} = 0.0254 \text{ mm.}$$

$$\delta L_2 = \frac{PL_2}{A_2 E} = \frac{200 \times 10^3 \times 0.16}{\frac{\pi}{4} \times (0.08)^2 \times 100 \times 10^9} = 0.0636 \text{ mm}$$

$$\delta L_3 = \frac{PL_3}{A_3 E} = \frac{200 \times 10^3 \times 0.12}{\frac{\pi}{4} \times (0.06)^2 \times 100 \times 10^9} = 0.0848 \text{ mm}$$

$\therefore$ Total elongation, $\delta L = \delta L_1 + \delta L_2 + \delta L_3 = 0.0254 + 0.0636 + 0.0848$
 $= 0.1738 \text{ mm}$

Problem 1.8 A steel bar of 30 mm diameter is loaded as shown in Fig 1.21. Determine the stresses in each part and the total elongation. Take $E = 210 \text{ GPa}$.

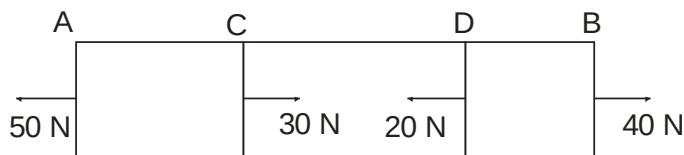


Fig. 1.21

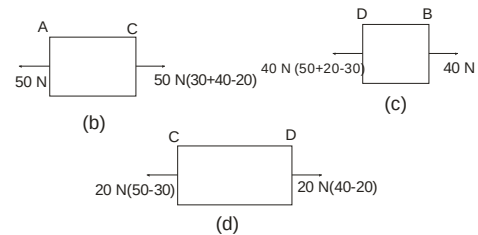


Fig. 1.22

Solution: The free body diagram for each portion is shown in Fig. 1.22.

Area of cross-section, $A = \frac{\pi}{4} \times (0.03)^2 = \pi \times 2.25 \times 10^{-4} \text{ m}^2$

$$\sigma_{AB} = \frac{50 \times 1000}{\pi \times 10^{-4} \times 2.25} = 70.77 \times 10^6 \text{ N/m}^2$$

$$\sigma_{BC} = \frac{20 \times 1000}{\pi \times 10^{-4} \times 2.25} = 28.31 \times 10^6 \text{ N/m}^2$$

$$\sigma_{CD} = \frac{40 \times 1000}{\pi \times 10^{-4} \times 2.25} = 56.61 \times 10^6 \text{ N/m}^2$$

$$\begin{aligned}
 \text{Total elongation , } \delta L &= \frac{P_1 L_1 + P_2 L_2 + P_3 L_3}{AE} \\
 &= \frac{50 \times 10^3 \times 0.4 + 20 \times 10^3 \times 0.3 + 40 \times 10^3 \times 0.2}{\pi \times 2.25 \times 10^{-4} \times 210 \times 10^9} \\
 &= 0.229 \text{ mm}
 \end{aligned}$$

Problem 1.9 If a tension bar is found to taper uniformly from $(D - a)$ mm diameter to $(D + a)$ mm, prove that the error involved in using the mean diameter to calculate young's Modulus is $\left(\frac{10a}{D}\right)^2$ percent.

Solution: we know that the elongation of a tapering rod is

$$\delta L = \frac{4PL}{\pi E D d}$$

Here $D_1 = (D + a)$ and $D_2 = (D - a)$

$$\therefore \delta L = \frac{4PL}{\pi E (D + a) (D - a)} = \frac{4PL}{\pi E (D^2 - a^2)}$$

$$\text{or } E = \frac{4PL}{\pi \delta L (D^2 - a^2)}$$

It is the actual value of young's modulus.

Let E_1 be the young's modulus calculated by taking mean diameter.

$$\text{Mean diameter} = \frac{(D + a) + (D - a)}{2} = D$$

$$\therefore E_1 = \frac{4PL}{\pi \delta L D^2} \quad [\because d = D]$$

$$\therefore \text{Percentage error} = \frac{E - E_1}{E} \times 100 = \frac{\frac{4PL}{\pi \delta L (D^2 - a^2)} - \frac{4PL}{\pi \delta L D^2}}{\frac{4PL}{\pi \delta L (D^2 - a^2)}} \times 100$$

$$= \frac{\frac{1}{D^2 - a^2} - \frac{1}{D^2}}{\frac{1}{D^2 - a^2}} \times 100 = \left(1 - \frac{D^2 - a^2}{D^2} \right) \times 100$$

$$= \frac{100a^2}{D^2} = \left(\frac{10a}{D} \right)^2 \text{ Proved.}$$

Problem 1.10 A rod of square section of side D at one end tapers to a square section of side d at the other end. The rod is of length l and is subjected to an axial pull P . Find the increase in length of the rod.

Solution: Consider a section AB at a distance x from the smaller end (Fig. 1.23)

$$AB = d + \frac{D-d}{l}x$$

$$\text{Cross-sectional area at } AB = \left[d + \frac{D-d}{l}x \right]^2$$

The elongation in length dx

$$= \frac{p \cdot dx}{\left[d + \frac{D-d}{l}x \right]^2 \times E}$$

The increase in length of the rod,

$$\begin{aligned} \delta L &= \int_0^L \frac{P dx}{\left(d + \frac{D-d}{l}x \right)^2 \times E} = \frac{P}{E} \left(\frac{-l}{D-d} \right) \left[\frac{1}{d + \frac{D-d}{l}x} \right]_0^L \\ &= \frac{P}{E} \left(\frac{-l}{D-d} \right) \left(\frac{1}{D} - \frac{1}{d} \right) = \frac{Pl}{EDd} \end{aligned}$$

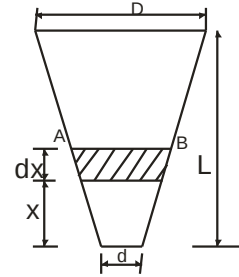


Fig. 1.23

Problem 1.11 A reinforced concrete column has a square cross-section. It is reinforced with 4 steel rods of diameter 20 mm at each corner. The column carries an axial compressive load of 1000 kN. Find the required size of the column so that the stress in the concrete does not exceed 4 N/mm^2 . Find also the load carried by steel and the stress induced in it. Assume modular ratio $\frac{E_s}{E_c} = 15$.

Solution: Given

Safe load, $P = 1000 \text{ kN}$, steel rod diameter, $d = 20 \text{ mm}$.

Number of steel rods = 4, stress in the concrete, $\sigma_c = 4 \text{ N/mm}^2$

$$\text{Area of steel rods } A_s = 4 \times \frac{\pi}{4} \times (20)^2 = 1256 \text{ mm}^2$$

Let the side of the column be $= x$

Area of the concrete $= A_c$

$$\text{Now } \frac{E_s}{E_c} = \frac{\sigma_s}{\sigma_c} \text{ or, } \frac{\sigma_s}{\sigma_c} = 15 \quad \therefore \sigma_s = 15 \sigma_c$$

Here $\sigma_C = 4 \text{ N/mm}^2$
 $\therefore \sigma_S = 15 \times 4 = 60 \text{ N/mm}^2$
 $\therefore$ Safe load, $P = P_S + P_C = \sigma_S \cdot A_S + \sigma_C \cdot A_C$
 $1000 \times 10^3 = 60 \times 1256 + 4 \times A_C$
 $\therefore A_C = 231160 \text{ mm}^2$
 Now, Total cross-sectional area,
 $x^2 = A_S + A_C = 1256 + 231160$
 $= 232416 \text{ mm}^2$
 $\therefore x = 482.1 \text{ mm}$
 Load carried by steel $= A_S \times \sigma_S = 1256 \times 60$
 $= 75360 \text{ N} = 75.36 \text{ kN}$
 Size of the column $= 485 \times 485 \text{ mm}$.

Problem 1.12 A solid steel bar 500 mm long and 70 mm in diameter is placed inside an aluminium tube having 75 mm inside diameter and 100 mm outside diameter. The aluminium tube is 0.15 mm longer than the steel bar. An axial load of 600 kN is applied to the bar and tube through rigid cover plates. Find the stresses developed in the steel bar and aluminium tube. E for steel $= 2.2 \times 10^5 \text{ N/mm}^2$, E for aluminium $= 0.7 \times 10^5 \text{ N/mm}^2$.

Solution:

$$A_S = \frac{\pi}{4} (70)^2 = 3846.5 \text{ mm}^2$$

$$A_{al} = \frac{\pi}{4} [(100)^2 - (75)^2] = 3434.4 \text{ mm}^2$$

Let the aluminium tube be compressed by δL_{al} and steel bar by δL_S

$$\delta L_{al} = \frac{\sigma_{al}}{E_{al}} \times L_{al}, \delta L_S = \frac{\sigma_S}{E_S} \times L_S$$

$$\frac{\sigma_{al}}{E_{al}} \times L_{al} - \frac{\sigma_S}{E_S} \times L_S = 0.15$$

$$\text{or } \frac{\sigma_{al}}{0.70 \times 10^5} \times 500.15 - \frac{\sigma_S}{2.20 \times 10^5} \times 500 = 0.15$$

$$\text{or } \frac{\sigma_{al}}{70} - \frac{\sigma_S}{220} = 300000$$

$$\sigma_{al} \cdot A_{al} + \sigma_S \cdot A_S = 600 \times 10^3$$

$$\text{or } \sigma_{al} \times 3434.4 + \sigma_S \times 3846.5 = 600 \times 10^3$$

$$\text{or } \sigma_{al} = 174.7 - 1.12 \sigma_S$$

Substituting the value of σ_{al} in Equation (i)

$$\frac{174.7 - 1.1199 \sigma_s}{70} - \frac{\sigma_s}{220} = 300000$$

$$\therefore \sigma_s = 106.82 \text{ N/mm}^2$$

$$\sigma_{al} = 54.99 \text{ N/mm}^2$$

Problem 1.13 Two copper rods and one steel rod together support a load 500 kN as shown in Fig. 1.24. Find the stresses in the rods. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$ and $E_c = 1 \times 10^5 \text{ N/mm}^2$.

Solution: Each rod will be compressed by the same amount.

Let the decrease in length of each rod be δ mm

Let the strain in copper and steel be e_c and e_s respectively

$$\therefore e_c l_c = e_s l_s = \delta$$

$$\therefore e_s = \frac{l_c}{l_s} e_c = e_c = \frac{140}{210} e_c$$

$$\therefore e_s = \frac{2}{3} e_c$$

Let the stress in steel and copper be σ_s and σ_c respectively.

$$\therefore \sigma_s = e_s E_s \text{ and } \sigma_c = e_c E_c$$

$$\therefore \frac{\sigma_s}{\sigma_c} = \frac{e_s}{e_c} \frac{E_s}{E_c}$$

$$\therefore \frac{\sigma_s}{\sigma_c} = \frac{2}{3} \times 2 = \frac{4}{3}$$

$$\therefore \sigma_s = \frac{4}{3} \sigma_c$$

Load on steel + Load on copper = Total load applied

$$\text{i.e., } \sigma_s A_s + \sigma_c A_c = P$$

$$\therefore \frac{4}{3} \sigma_c \times 3200 + \sigma_c \times (2 \times 2000) = 250000$$

$$\therefore \frac{128}{3} \sigma_c + 40 \sigma_c = 2500$$

$$\therefore \frac{248}{3} \sigma_c = 2500$$

$$\therefore \sigma_c = 30.24 \text{ N/mm}^2$$

$$\text{And } \sigma_s = \frac{4}{3} \times 30.24 = 40.32 \text{ N/mm}^2$$

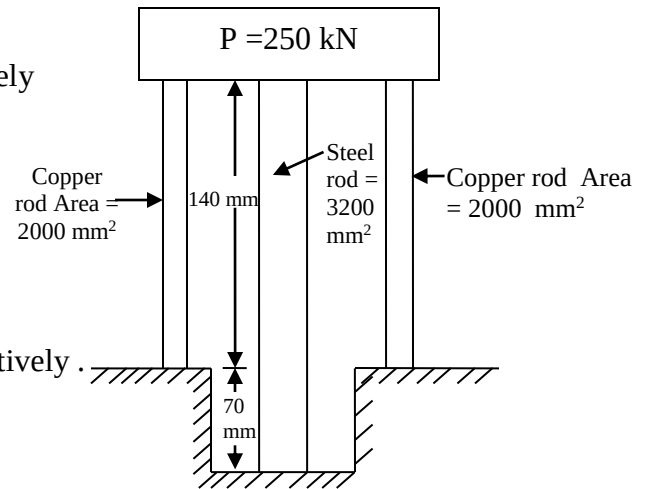


Fig. 1.24

.....(1.36)

Problem 1.14 A steel rod 18 mm in diameter passes centrally through a steel tube 25 mm in internal diameter and 30 mm in external diameter. The tube is 700 mm long and is closed by rigid washers of negligible thickness which are fastened by nuts threaded on the rod. The nuts are tightened until the compressive load on the tube is 15 kN. Calculate the stresses in the tube and the rod.

Find the increase in these stresses when one nut is tightened by one quarter of a turn relative to the other. There are 4 threads per 10 mm. Take $E = 2 \times 10^5 \text{ kN/mm}^2$

Solution: When the nuts are tightened the tube will be compressed and the rod will be elongated. Since no external forces have been applied, the compressive load on the tube must be equal to the tensile load on the rod.

Area of the tube

$$A_t = \frac{\pi}{4} (30^2 - 25^2) \text{ mm}^2 = \frac{275\pi}{4} \text{ mm}^2$$

Area of the rod

$$A_r = \frac{\pi}{4} \times 18^2 = 81\pi \text{ mm}^2$$

Let the stresses in the rod and tube be p_r and p_t N/mm² respectively.

Tensile load on the rod = compressive load on the tube

$\therefore$

$$\sigma_r A_r = \sigma_t \cdot A_t$$

$\therefore$

$$\sigma_r = \frac{A_t}{A_r} \cdot \sigma_t$$

$\therefore$

$$\sigma_r = \frac{275\pi}{4} \cdot \frac{1}{81\pi} \sigma_t$$

$\therefore$

$$\sigma_r = \frac{275}{324} \sigma_t$$

(I) when the compressive load on the tube is 15 kN

$$\therefore \text{Stress in the tube } \sigma_t = \frac{15000}{\left(\frac{275\pi}{4}\right)} = 69.5 \text{ N/mm}^2 \text{ (Compressive)}$$

$$\therefore \text{Stress in the rod } \sigma_r = \frac{275}{324} \times 69.5 \text{ N/mm}^2 = 58.99 \text{ N/mm}^2 \text{ (tensile)}$$

(ii) when one nut is tightened by one quarter of a turn.

Let σ_t and σ_r be the stresses due to tightening of the nut by one-quarter of a turn.

Obviously σ_t is a compressive stress and σ_r is tensile, and

$$\sigma_r = \frac{275}{324} \sigma_t$$

$\therefore$ Reduction in the length of the tube

$$= \frac{\sigma_t}{E} l = \frac{\sigma_t \times 700}{200000} \text{ mm} = 0.00350 \sigma_t \text{ mm}$$

Extension of the rod

$$\begin{aligned} &= \frac{\sigma_r}{E} l = \frac{\sigma_r \times 700}{200000} \text{ mm} \\ &= \frac{275}{324} \cdot \frac{\sigma_t \times 700}{200000} \text{ mm} = 0.002971 \sigma_t \text{ mm} \end{aligned}$$

But contraction of the tube + extension of the rod = axial advance of the nut.

$$0.00350 \sigma_t + 0.002971 \sigma_t = \frac{1}{4} \cdot \frac{1}{4} (10) \text{ mm} = 0.625 \text{ mm}$$

$$\therefore 0.006471 \sigma_t = 0.625$$

$$\therefore \sigma_t = 96.6 \text{ N/mm}^2 \text{ (compressive)}$$

$$\sigma_r = \frac{275}{324} \times 96.6 = 81.98 \text{ N/mm}^2 \text{ (tensile)}$$

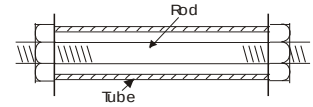


Fig. 1.25

Problem 1.15 A rod is 2 meters long at 15°C . Find the expansion of the rod when the temperature is raised to 85°C . If this expansion is prevented, find the stress in the material. Take $E = 1 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012 \text{ per } ^\circ \text{C}$.

Solution: Rise in temperature = $T = 85 - 15 = 70^\circ \text{C}$

$$\text{Free expansion} = \alpha T l = 0.000012 \times 70 \times 2 \times 1000 = 1.68 \text{ mm}$$

$$\text{Temperature stress} = \alpha T E = 0.000012 \times 70 \times 1 \times 10^5 = 84 \text{ N/mm}^2$$

Problem 1.19 A rod of steel is 20 meters long at a temperature of 30°C . Find the free expansion of the rod when the temperature is raised to 75°C . Find the temperature stress produced (i) When the expansion of the rod is prevented (ii) when the rod is permitted to expand by 6.8 mm. Take $\alpha = 12 \times 10^{-6} \text{ per } ^\circ \text{C}$ and $E = 2 \times 10^5 \text{ N/mm}^2$

Solution:

$$\text{Free expansion of the rod} = \alpha T l = 12 \times 10^{-6} (75 - 30) 20 \times 2 \times 1000 \text{ mm} = 10.80 \text{ mm}$$

(i) When the expansion is fully prevented

$$\begin{aligned} \text{Temperature stress} &= \alpha T E = 12 \times 10^{-6} \times (75 - 30) 2 \times 10^5 \text{ N/mm}^2 \\ &= 108 \text{ N/mm}^2 \end{aligned}$$

(ii) when the rod is permitted to expand by 6.8 mm

In this case, expansion prevented = 10.8 - 6.8 = 4mm

$$\therefore \text{Strain} = \frac{\text{Expansion prevented}}{\text{Original Length}} = \frac{4}{20 \times 1000} = \frac{3}{5000}$$

$$\therefore \text{Temperature stress} = \text{Strain} \times E = \frac{1}{5000} \times 2 \times 10^5 \text{ N/mm}^2 = 40 \text{ N/mm}^2$$

Problem 1.16 A 15 mm diameter steel rod passes centrally through a copper tube 60 mm external diameter and 50 mm internal diameter. The tube is closed at each end by rigid plates of negligible thickness. The nuts are tightened lightly on the projecting parts of the rod. If the temperature of the assembly is raised by 60°C, calculate the stresses developed in copper and steel.

$$\begin{aligned} \text{Take } E_s &= 2.10 \times 10^5 \text{ N/mm}^2; E_c = 1.05 \times 10^5 \text{ N/mm}^2; \\ \alpha_s &= 12 \times 10^{-6} \text{ per } ^\circ\text{C}; \alpha_c = 17.5 \times 10^{-6} \text{ per } ^\circ\text{C} \end{aligned}$$

Solution: Area of the steel rod $= A_s = \frac{\pi}{4} (15)^2 \text{ mm}^2 = 56.25 \pi \text{ mm}^2$

Area of the copper tube $= A_c = \frac{\pi}{4} (60^2 - 50^2) \text{ mm}^2 = 275 \pi \text{ mm}^2$

Free expansion of steel = $\alpha_s T l$

Free expansion of copper = $\alpha_c T l$

Let the actual expansion of each component be δ

$$\alpha_c T l > \delta > \alpha_s T l$$

$\therefore$ Steel is in tension and copper is in compression

Let σ_s and σ_c be the stresses in steel and copper.

For the equilibrium of the system

Tension in steel = Compression in copper

$$\sigma_s A_s = \sigma_c A_c$$

$$\sigma_s = \left[\frac{A_c}{A_s} \right] \sigma_c$$

$$\therefore \sigma_s = \left[\frac{275\pi}{56.25\pi} \right] \sigma_c$$

$$\therefore \sigma_s = 4.89 \sigma_c$$

Actual expansion of steel = Actual expansion of copper

$$\alpha_s T l + \frac{\sigma_s}{E} l = \alpha_c T l - \frac{\sigma_c}{E_c} l$$

$$\therefore \alpha_s T + \frac{\sigma_s}{E_s} = \alpha_c T - \frac{\sigma_c}{E_c}$$

But $\sigma_s = 4 \sigma_c$ and substituting for α_s , α_c , E_s and E_c , we get

$$12 \times 10^{-6} \times 60 + \frac{4.89 \sigma_c}{2.10 \times 10^5} = 17.5 \times 10^{-6} \times 60 - \frac{\sigma_c}{1.05 \times 10^5}$$

$$72 + 2.32 \sigma_c = 105 - 0.952 \sigma_c$$

$$3.28 \sigma_c = 33$$

$$\sigma_c = 10.06 \text{ N/mm}^2 \text{ (compressive)}$$

$$\sigma_s = 4 \times 10.06 = 40.24 \text{ N/mm}^2 \text{ (tensile)}$$

Problem 1.17 A copper rod 22 mm diameter screwed at the ends passes through a steel tube 25 mm and 30 mm internal and external diameters. The temperature of the whole assembly is raised to 116°C and the nuts on the rod are then screwed lightly on the ends of the tube. Find the intensity of stress in the rod and the tube when the common temperature has fallen to 16°C .

$$\text{Coefficient of expansion for steel} = 12 \times 10^{-6} \text{ per } ^\circ \text{C}$$

$$\text{Coefficient of expansion for copper} = 17.5 \times 10^{-6} \text{ per } ^\circ \text{C}$$

$$\text{Modulus of Elasticity for steel} = 2.1 \times 10^5 \text{ N/mm}^2$$

$$\text{Modulus of Elasticity for copper} = 1.05 \times 10^5 \text{ N/mm}^2$$

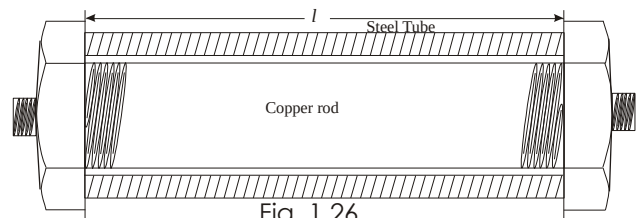
$$\text{Area of the steel tube } A = \frac{\pi}{4} [30^2 - 25^2] = \frac{275}{4} \pi \text{ mm}^2$$

Solution:

$$\text{Area of the gun metal rod } A_s = \frac{\pi}{4} \times 22^2 = 121\pi \text{ mm}^2$$

Let the length of the rod and the tube between the nuts be l mm. If the two members had been free to contract, free contraction of the copper rod $= \alpha_c Tl$, free contraction of the steel tube $= \alpha_s Tl$

Since α_c is greater than α_s the free contraction of the copper rod is greater than the free contraction of the steel tube. But, since the ends of the rod have been provided with nuts the two members are not free to contract fully, each of the member will contract by the same amount. Let δ mm be the final contraction of the each rod. The free



contraction of the copper rod is greater than δ , while the free contraction of the steel tube is less than δ . Hence the steel tube will be subjected to compressive stress while the copper rod will be subjected to tensile stress. Let σ_s and σ_c be the stress in steel and copper.

For the equilibrium of the whole system:

Total compressive force in steel = Total tension in gun metal.

$$\begin{aligned}\therefore \quad \sigma_s A_s &= \sigma_c A_g \\ \sigma_s &= \frac{A_g}{A_s} \sigma_c = \frac{121\pi}{\left(\frac{25\pi}{4}\right)} \sigma_c\end{aligned}$$

$$\therefore \quad \sigma_s = 1.76 \sigma_c$$

Final contraction of steel = final contraction of gun metal

$$\therefore \quad \alpha_s T l + \frac{\sigma_s}{E_s} l = \alpha_g T l - \frac{\sigma_c}{E_g} l$$

$$\therefore \quad \alpha_s T + \frac{\sigma_s}{E_s} = \alpha_g T - \frac{\sigma_c}{E_g}$$

$$\text{Since } T = 126^\circ - 16^\circ = 110^\circ \text{ C}$$

$$\therefore \quad 12 \times 10^{-6} \times 110 + \frac{1.76 \sigma_c}{2.1 \times 10^5} = 17.5 \times 10^{-6} \times 110 - \frac{\sigma_c}{1.05 \times 10^5}$$

$$132 + 0.838 \sigma_c = 192.5 - 0.952 \sigma_c$$

$$1.79 \sigma_c = 60$$

$$\sigma_c = 33.8 \text{ N/mm}^2 \text{ (Tensile)}$$

$$\sigma_s = 1.76 \times 33.8 = 59.5 \text{ N/mm}^2 \text{ (Compressive)}$$

Problem 1.18 A steel bar is placed between two copper bars each having the same area and length as the steel bar at 20° C . At this stage they are rigidly connected together at

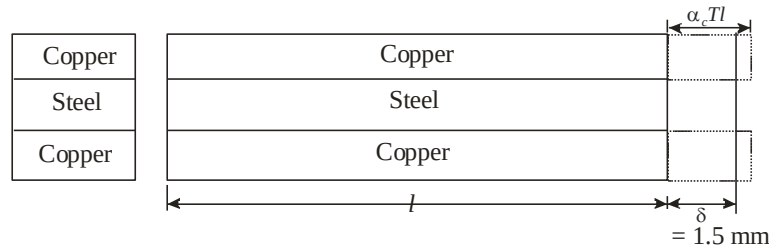


Fig. 1.27

both the ends. When the temperature is raised to 320° C , the length of the bars increases by 1.50 mm. Determine the original length and the final stresses in the bars.

Take $E_s = 2.1 \times 10^5 \text{ N/mm}^2$; $E_c = 1 \times 10^5 \text{ N/mm}^2$

$$\alpha_s = 0.000012 \text{ per } ^\circ\text{C}$$

$$\alpha_c = 0.1111175 \text{ per } ^\circ\text{C}$$

Solution: Let the sectional area of the steel component be $A \text{ mm}^2$

$\therefore$ Sectional area of the copper component = $2A \text{ mm}^2$

Free expansion of steel component = $\alpha_s Tl$

Free expansion of copper component = $\alpha_s Tl$

Let δ be the actual expansion of each bar.

$$\alpha_s Tl < \delta < \alpha_c Tl$$

$\therefore$ Steel is in tension and copper is in compression.

Let σ_s and σ_c be the stresses in steel and copper respectively.

For the equilibrium of the system.

Tension in steel = Compression in copper.

$$\therefore \sigma_s A = \sigma_c (2A)$$

$$\therefore \sigma_s = 2 \sigma_c$$

Actual expansion of steel = Actual expansion of copper.

$$\alpha_s Tl + \frac{\sigma_s}{E_s} l = \alpha_c Tl - \frac{\sigma_c}{E_c} l$$

$$\alpha_s T + \frac{\sigma_s}{E_s} = \alpha_c T - \frac{\sigma_c}{E_c}$$

$$T = 320 - 20 = 300^\circ \text{C}$$

$$\therefore 0.000012 \times 300 + \frac{2\sigma_c}{2.1 \times 10^5} = 0.0000175 \times 300 - \frac{\sigma_c}{1 \times 10^5}$$

$$360 + 0.952 \sigma_c = 525 \sigma_c$$

$$1.952 \sigma_c = 165$$

$$\therefore \sigma_c = 84.52 \text{ N/mm}^2 \text{ (compressive)}$$

$$\therefore \sigma_s = 2 \times 84.52 = 169.04 \text{ N/mm}^2 \text{ (tensile)}$$

Actual expansion of the steel bar = Actual expansion of the copper bar

$$= \alpha_s Tl + \frac{\sigma_s}{E_s} l = 1.50$$

$$0.000012 \times 300l + \frac{169.04}{2.1 \times 10^5} l = 1.50$$

$$360l + 80.495l = 150000$$

$$440.495l = 150000$$

$$l = 340.526 \text{ mm}$$

Problem 1.19 A steel bar is placed between two copper bars each having the same area and length as the steel bar at 15°C . At this stage they are rigidly connected together at both the ends. When the temperature is raised to 315°C , the length of the bars increases by 1.5 mm. Determine the original length and the final stresses in the bars.

Take $E_s = 2.1 \times 10^5 \text{ N/mm}^2$ $\alpha_s = 12 \times 10^{-6}/^{\circ}\text{C}$

$E_c = 1 \times 10^5 \text{ N/mm}^2$ $\alpha_c = 17.2 \times 10^{-6}/^{\circ}\text{C}$

Solution:

Let a = Area of steel bar
 2a = Area of copper bar

Free expansion of steel bar = $L \alpha_s T$

Free expansion of copper bar = $L \alpha_c T$

Actual expansion of each bar = δL

$\therefore L \alpha_s T < \delta L < L \alpha_c T$

$\therefore$ Steel bar is in tension and copper bar is in compression.

For equilibrium of the system

Tension in steel bar = Compression in copper bar

$\sigma_s a = \sigma_c (2a)$

$\therefore \sigma_s = 2 \sigma_c$

Actual expansion of steel = Actual expansion of copper

$$L \alpha_s T + \frac{\sigma_s}{E_s} \times L = L \alpha_c T - \frac{\sigma_c}{E_c} \times L$$

$$L \alpha_s T + \frac{\sigma_s}{E_s} = L \alpha_c T - \frac{\sigma_c}{E_c}$$

Hear $T = 315 - 15 = 300^{\circ}\text{C}$

$$12 \times 10^{-6} \times 300 + \frac{2\sigma_c}{2.1 \times 10^5} = 17.2 \times 10^{-6} \times 300 - \frac{\sigma_c}{1 \times 10^5}$$

$\therefore \sigma_c = 84.511 \text{ N/mm}^2$

$\sigma_s = 169.022 \text{ N/mm}^2$

Actual expansion of steel bar = $L \alpha_s T + \frac{\sigma_s}{E_s} \times L = 1.5$

$$L \left(12 \times 10^{-6} \times 300 + \frac{169.022}{2.1 \times 10^5} \right) = 1.5$$

$L \times 4.405 = 1.5$

$\therefore L = 340.5 \text{ mm}$

Problem 1.20 The composite bar consisting of steel and aluminium compound shown in fig is connected to two grips at the ends at a temperature of 60°C . Find the stress in the two rods when the temperature falls to 20°C (i) if the ends do not yield, (ii) if the ends

yield by 0.25 mm. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$, $E_a = 0.07 \times 10^5 \text{ N/mm}^2$, $\alpha_s = 1.17 \times 10^{-5} \text{ per } ^\circ\text{C}$ and $\alpha_a = 2.34 \times 10^{-5} \text{ per } ^\circ\text{C}$. Area of the steel and aluminium bars are 250 mm^2 and 400 mm^2 respectively.

Solution: $\frac{A_a}{A_s} = \frac{400}{250} = 1.6$

Free contraction of the composite bar

$$= \alpha_s T l_s + \alpha_a T l_a$$

$$= 1.17 \times 10^{-5} (60 - 20) 800 + 2.34 \times 10^{-5} (60 - 20) 400 = 0.7488 \text{ mm}$$

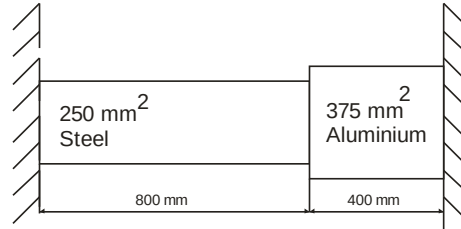


Fig. 1.28

When the contraction is prevented completely or partially tensile stress are induced in the rods. Let σ_s and σ_a be the stress in the steel and aluminium rods.

Since the same force exists in the two rods. $A_s \sigma_s = A_a \sigma_a \quad \therefore \sigma_s = \frac{A_a}{A_s} \sigma_a = \frac{400}{250} \sigma_a$

$$\therefore \sigma_s = 1.6 \sigma_a$$

Case (i) When the ends do not yield

Contraction prevented = 0.7488 mm

Contraction prevented in steel + contraction prevented in aluminium = 0.7488 mm.

$$\frac{P_s}{E_s} l_s + \frac{P_a}{E_a} l_a = 0.7488$$

$$\frac{1.6 \sigma_a}{2 \times 10^5} \times 800 + \frac{\sigma_a}{0.70 \times 10^5} \times 400 = 0.7488 \quad \therefore 12.1143 \times 10^{-3} \sigma_a = 0.7488$$

$$\sigma_a = 61.8 \text{ N/mm}^2$$

$$\sigma_s = 1.6 \times 61.8 = 98.88 \text{ N/mm}^2$$

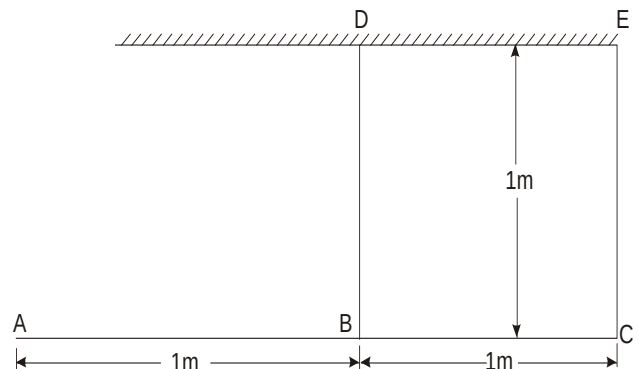
Case (ii) When the ends yield by 0.25 mm.

$$\text{Contraction prevented} = 0.7488 - 0.25 = 0.4988 \text{ mm}$$

$$12.1143 \times 10^{-3} \sigma_a = 0.4988 \text{ mm} \quad \therefore \sigma_a = 41.17 \text{ N/mm}^2$$

$$\therefore \sigma_s = 1.6 \times 41.17 = 65.88 \text{ N/mm}^2.$$

Problem 1.21 Fig. 1.29 shows a rigid bar ABC hinged at A and suspended at two points B and C by two bars BD and CE made of aluminium and steel respectively. The bar carries a load of 20 kN midway between B and C. The cross-sectional area of the aluminium bar BD is 4 sq. mm and that of the steel bar CE



is 3 sq. mm. Determine the load taken by the two bars BD and CE.

Modulus of Elasticity for aluminium $E_a = 7 \times 10^4 \text{ N/mm}^2$

Modulus of Elasticity for steel $E_s = 2 \times 10^5 \text{ N/mm}^2$

Solution: A rigid rod will not bend. It will remain straight. But the aluminium and steel rods will extend due to their elasticity. Fig shows the position of the rigid bar after the aluminium and steel rods have undergone their extensions.

Let the tensions in the aluminium and steel rods be P Newton and Q Newton respectively.

Since the rigid bar remains straight, the extensions of CE and BD are proportional to their distances from A.

Let the extension of BD be $BB_1 = \Delta$

$$\therefore \text{Extension of CE} = CC_1 = \frac{2}{1} \times \Delta = 2\Delta$$

$$\text{But extension of BD} = \Delta = \frac{P}{A_a E_a}$$

$$\text{And extension of CE} = 2\Delta = \frac{Q}{A_s E_s}$$

$$\therefore \Delta = \frac{Pl}{A_a E_a} = \frac{Ql}{2A_s E_s}$$

$$\therefore \frac{P}{Q} = \frac{1}{2} \cdot \frac{A_a}{A_s} \cdot \frac{E_a}{E_s}$$

$$\therefore \frac{P}{Q} = \frac{1}{2} \cdot \frac{4}{3} \cdot \frac{70000}{200000}$$

$$\therefore \frac{P}{Q} = \frac{28}{120}$$

$$\therefore P = \frac{28}{120} Q \quad \dots\dots(1.37)$$

Now consider the equilibrium of the rigid bar.

Taking moments of the forces of the rigid bar, about the end A, we have.

$$P \times 1 + Q \times 2 = 20,000 \times 1.5$$

$$\therefore P + 2Q = 30000 \quad \dots\dots(1.38)$$

$$\text{But } P = \frac{28}{120} Q \quad \dots\dots(1.39)$$

$$\therefore \frac{28}{120} Q + 2Q = 30000$$

$$\therefore \frac{268}{120} Q = 30,000$$

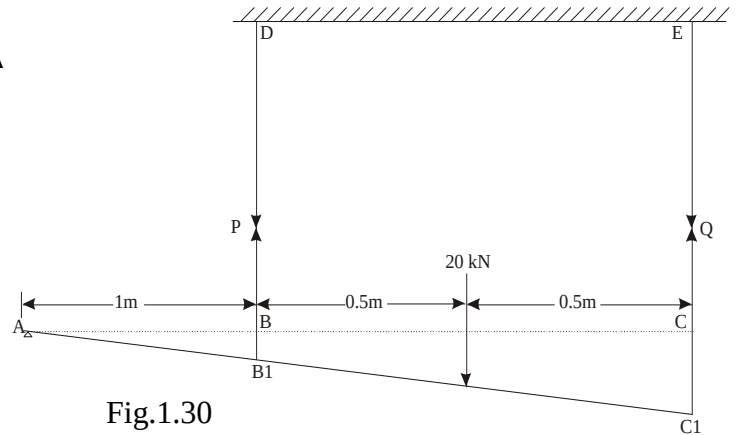
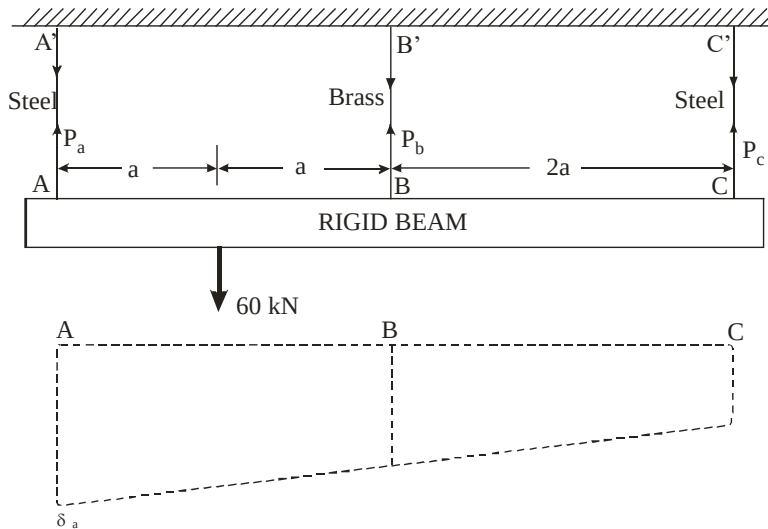


Fig.1.30

$$\therefore Q = \frac{30000 \times 120}{268} = 13432.8 \text{ N (tensile)}$$

$$\therefore P = \frac{28}{120} \times 13260 = 3134 \text{ N (tensile)}$$

Problem 1.22 A rigid beam carrying a load 60 kN is supported by three bars as shown in fig. All the bars are of the same area A and same length l . Find the load carried by each bar. Neglect the weight of the beam. Take $E_s = 2E_b$.



Solution: Let P_a , P_b and P_c be the tensions in the bars AA' , BB' and CC' respectively. Consider the equilibrium of the beam.

Resolving moments about C,

$$P_a \times 4a + P_b \times 2a = 60 \times 3a$$

$$\therefore 2P_a + P_b = 90$$

Let extensions of AA' , BB' and CC' be Δ_a , Δ_b and Δ_c respectively. Obviously Δ_a , Δ_b and Δ_c are the deflections of the beam at A, B and C.

$$\Delta_a = \frac{P_a l}{AE_s}, \Delta_b = \frac{P_b l}{AE_b} \text{ and } \Delta_c = \frac{P_c l}{AE_s}$$

Since the beam is rigid, it remains straight

$$\therefore \Delta_b = \frac{\Delta_a + \Delta_c}{2}$$

$$\therefore \frac{P_b l}{AE_b} = \frac{1}{2} \left[\frac{P_a l}{AE_s} + \frac{P_c l}{AE_s} \right] = \frac{l}{2AE_s} [P_a + P_c] = \frac{l}{4AE_b} [P_a + P_c]$$

$$\therefore P_b = \frac{P_a + P_c}{4}$$

$$\therefore P_a + P_c = 4P_b \quad \dots(1.40)$$

Solving equations (i), (ii) and (iii), we get,

$$P_a = 39 \text{ kN}, P_b = 12 \text{ kN} \text{ and } P_c = 9 \text{ kN}$$

Exercise

What is the procedure of finding thermal stresses in the composite bars?

1.1 Find an expression for the total elongation of a uniformly tapering rectangular bar when it is subjected to an axial load P .

1.2 Find the Young's modulus of a rod of diameter 30 mm and of length 300 mm which is subjected to a tensile load of 60 kN and the extension of the rod is 0.4 mm.

(Ans. 63.5 GN/m²)

1.4 An axial pull of 40 kN is acting on a bar consisting of three sections of length 300 mm, 250 mm and 200 mm and of diameters 20 mm, 40 mm and 50 mm respectively. If the Young's modulus is 2×10^5 N/mm² determine (a) stresses in each section (b) total extension of the bar.

(Ans. 127.32, 31.8, 20.37 N/mm², 0.25 mm)

1.5 A rod which tapers uniformly from 50 mm diameter to 30 mm diameter in a length of 500 mm, is subjected to an axial load of 6 kN. If the Young's modulus is 2×10^5 N/mm² find the extension of the rod.

(Ans. 0.0127 mm)

1.6 A rectangular bar of made of steel is 3 m long and 10 mm thick. The bar is subjected to an axial tensile pull of 50 kN. The width of the bar varies from 70 mm at one end to 28 mm at the other. If the Young's modulus is 2×10^5 N/mm² find the extension of the rod.

(Ans. 1.636 mm)

1.7 Find the Young's modulus for a rod which tapers uniformly from 40 mm diameter to 25 mm diameter in a length of 400 mm. The rod is subjected to a load of 6 kN and extension of the rod is 0.04 mm.

(Ans. 76.39 kN/mm²)

1.8 A reinforced concrete column 250 mm x 250 mm in section is reinforced with 8 steel bars. The total area of steel bars is 1608.5 mm². The column carries a load of 270 kN. If the Young's modulus for steel is 18 times that of concrete, find the stresses in concrete and steel. If the stress in concrete shall not exceed 4 N/mm² find the area of steel required so that the column may support a load of 400 kN.

(Ans. $\sigma_c = 3$ N/mm², $\sigma_s = 54$ N/mm², $A_s = 2206$ mm²)

1.9 A vertical bar fixed at upper end and of uniform strength carries a tensile load of 500 kN. The bar is 18 m long and having weight per unit volume as 0.00008 N/mm^2 . If the area of the bar at the lower end is 500 mm^2 find the area of the bar at the upper end.

(Ans. 500.72 mm^2)

1.10 A rod is 3 m long at a temperature of 15°C . Find the expansion of the rod when the temperature is raised to 95°C . If this expansion is prevented find the stress induced in the material of the rod. If the Young's modulus is $1 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012 / ^\circ\text{C}$

(Ans. 2.88 mm, 96 N/mm^2)